GNITS-R- 18 –117FG

G. Narayanamma Institute of Technology & Science

(Autonomous) ***(for Women)***

Shaikpet, Hyderabad- 500 104

IV-B.Tech I-Semester *Regular / Supplementary Examinations, Nov* - 2024

DATA SCIENCE USING R

(Computer Science and Engineering (Data Science))

Max. Marks: 70 **Time: 03 Hours**

Note:

1. Question paper comprises of **Part A** and **Part B**.
2. **Part A** is compulsory which carries 10 marks. Answer all questions in Part A.
3. **Part B** (for 60 marks)consists of **five questions** with **"either" "or"** pattern. Each question carries 12 marks and may have a,b,c as sub questions. The student has to answer any one full question.

PART-A

*(Answer 05 questions. Each question carries **2 marks**)*

Q.No.	Question	Marks	CO	BTL
Q.1	a) What is Data Science and mention any two of its application areas?	**[02]**	**CO1**	**[L1]**
	b) Describe any two R-functions that will be used to handle Data in Data Frames.	**[02]**	**CO1**	**[L2]**
	c) Mention any two ways of validating models.	**[02]**	**CO2**	**[L1]**
	d) Define Hierarchical clustering and mention its use.	**[02]**	**CO3**	**[L2]**
	e) What is seasonal adjusting? Give an example.	**[02]**	**CO5**	**[L2]**

END OF PART A

PART-B

*(Answer **05** full questions. Each question carries **12 marks**)*

Q.No.	Question	Marks	CO	BTL
Q.2(a)	Describe the various stages involved in a data science project by emphasizing the importance of each stage.	**[06]**	**CO2**	**[L2]**
(b)	Explain the concept of working with directories in R and why they are essential?	**[06]**	**CO1**	**[L2]**
	OR			
Q.3(a)	Explain the process of loading and handling data in R.	**[06]**	**CO1**	**[L2]**
(b)	Write short note on the common data types in R for representing information.	**[06]**	**CO1**	**[L2]**

Q.4(a)	Illustrate the process of loading a data frame in R with an example code.	**[06]**	**CO1**	**[L2]**
(b)	What is the significance of data summary statistics in data analysis, and how can they be generated in R?	**[06]**	**CO2**	**[L1]**
	OR			
Q.5(a)	"Data visualization help in identifying and addressing data problems"- Justify this statement with suitable justification.	**[06]**	**CO4**	**[L3]**
(b)	Define Outlier. Explain the procedure of identifying outliers in a data frame using R.	**[06]**	**CO3**	**[L1]**
Q.6(a)	Describe the procedure of building single-variable models with suitable example.	**[06]**	**CO3**	**[L2]**
(b)	What does KDD stand for in the context of machine learning, and write short note on the contributions of KDD Cup 2009?	**[06]**	**CO6**	**[L1]**
	OR			
Q.7(a)	What is the purpose of evaluating models in machine learning? Briefly explain.	**[06]**	**CO3**	**[L2]**
(b)	Briefly discuss the key advantage of building models using many variables in machine learning.	**[06]**	**CO3**	**[L3]**
Q.8(a)	Write short note on the primary goal of cluster analysis in data mining.	**[06]**	**CO4**	**[L2]**
(b)	How does hierarchical clustering work, and what is the main advantage of this method in cluster analysis?	**[06]**	**CO6**	**[L3]**
	OR			
Q.9(a)	What is logistic regression, and what type of outcomes does it predict?	**[06]**	**CO4**	**[L1]**
(b)	What is the key principle behind the k-means algorithm, and how is the number of clusters determined in this approach?	**[06]**	**CO6**	**[L4]**
Q.10(a)	Write short note on the basic need of decomposing time series data. Justify with an example application.	**[06]**	**CO5**	**[L2]**
(b)	Describe the steps involved in decomposing time series data into its seasonal and non-seasonal components.	**[06]**	**CO5**	**[L2]**
	OR			
Q.11(a)	Discuss the basic principles and applications of regression and trend analysis in understanding and modeling time-dependent patterns.	**[06]**	**CO5**	**[L2]**
(b)	What is the goal of similarity search in time series data analysis? Briefly explain.	**[06]**	**CO5**	**[L1]**

END OF PART B

END OF THE QUESTION PAPER